

Soil water repellency under Eucalyptus stands of different age and its relationship with soil hydrological function and soil organic matter

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ABSTRACT

Eucalyptus stands can modify soil hydrological functioning, by several mechanisms including the effect of the growing root system, changes in soil organic matter (SOM) and the occurrence of soil water repellency (SWR). This effect is expected to evolve with time, as tree root system develop and SOM is formed from tree litter. SWR can prevent water from entering the soil, reducing infiltration, while the effect of roots and SOM can improve structure, favoring infiltration. The aims of this study were: i- to determine the influence of Eucalyptus stands of different ages on soil physical and hydraulic properties and SWR phenomena; ii- to evaluate the effects of SOM content and composition on SWR; iii- to determine the relationship between SWR and soil physical properties; iv- to compare different methodologies for SWR evaluation. The soil studied was a in a silty loam Typic Argiudoll with Eucalyptus stands of 3, 7, 11 and 32 years. Samples were taken to determine saturated and near saturation hydraulic conductivity, water retention curve, total organic carbon and different fractions of SOM, and SWR. Different parameters were used to estimate SWR, including the repellency index (RI), water repellency cessation time (WRCT), modified repellency index (Rim), and a qualitative analysis of the shapes of cumulative infiltration curves. As main results, a reduction in total porosity (TP) in the first years of Eucalyptus plantation was found, followed by a steep increase in TP, mainly due to an increase in microporosity. Hydraulic conductivity decreased with stand age, which was attributed to a decrease in macro and mesoporosity and porosity connectivity. SOM and most SOM fractions were not affected by stand age. RI was a consistent method to estimate SWR in Eucalyptus stands, while WRCT and Rim determination was more difficult when non classical shapes of infiltration curves were obtained.

1. Introduction

Soil water repellency (SWR), which is caused by the coating of mineral components by organic compounds (Majid et al., 2023), is a phenomena that affects most soils in different degree (Li et al., 2023)

and that can alter soil hydrological functioning. Although SWR is widespread and can significantly change water dynamics, it is still not well understood and is often not considered in hydrological models (Di Prima et al., 2021; Li et al., 2023). SWR can reduce infiltration rates in the initial steps of infiltration (Alagna et al., 2017), therefore favoring

Abbreviations: SWR, Soil water repellency; SOM, soil organic matter; K(h), hydraulic conductivity; Se, ethanol sorptivity; Sw, water sorptivity; WRCT, water repellency cessation time; RI, repellency index; Rim, modified repellency index; TP, total porosity; SS, specific surface; TOC, total organic carbon; h, tension; θ_{ma} , θ_{me} , θ_{mi} , macro, meso and microporosity; PAW, plant available water capacity; FC, field capacity; P_{tr} , transmission porosity; P_{st} , storage porosity, K_0 , K_3 , K_6 , hydraulic conductivity at 0, -3, and -6 cm h; Cw_{TP} , total porosity connectivity; FA, fulvic acids; HA, humic acids; hum, humins; POCc, coarse particulate organic carbon; POCf, fine particulate organic carbon; MOC, mineral associated organic carbon.

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water runoff and soil erosion (Holko et al., 2023). SWR can cause irregular water fronts and enhance preferential flow (Rye & Smettem, 2017; Smettem et al., 2021), increasing nutrient and pollutants leaching (Jarvis, 2007). Also, it has been suggested that SWR may increase water availability by reducing water potential at a certain moisture content (Goebel et al., 2011; Smettem et al., 2021), and affect the drying process by reducing evaporation from the soil (Rye & Smettem, 2017; Li et al., 2023). Other authors state that SWR may prevent soil organic matter (SOM) decomposition, but can also increase runoff events and thus soil degradation (Goebel et al., 2011). Furthermore, SWR depends on drying and wetting cycles, and can be aggravated under drought conditions (Li et al., 2023), increasing flooding risk if these periods are alternated with high intensity rainfalls, as can be expected to happen more often as a consequence of climate change (IPCC, 2022; Scherger et al., 2022).

Among other things, SWR depends on plant cover, with certain species of trees being more prone to trigger SWR. It has been reported that Eucalyptus cause SWR in a higher degree than other tree species such as oak and pine trees (Zavala et al., 2014; Majid et al., 2023). Eucalyptus plantations are worldwide spread for industrial purposes, being the second most cultivated genus and representing 26 % of implanted forests (Binkley et al., 2017) and around 22 million hectares worldwide (Hoogar et al., 2019). SWR in Eucalyptus forests may be caused by changes in the amount and composition of SOM, resulting from organic material decomposition and leaching of organic substances such as root exudates (e.g. resins and waxes), and changes in the associated micro biota (Doerr et al., 2000; Majid et al., 2023). Also, tree plantations can alter soil hydraulic properties by different mechanisms including changes in soil aggregation and soil pore system configuration (Buczko et al. 2002; Alagna et al., 2017; Di Prima et al., 2023). Stand age may be a key factor governing the influence of forests in soil hydrological properties and SWR, as tree growing process affects soil physical and chemical properties (Zhao et al., 2022; Zema et al., 2021). However, little literature exists about the effect of stand age on SWR, $K(h)$ and other physical properties (Julich et al., 2021; Zema et al., 2021). Zhao et al. (2023) studied the first 5 years of Eucalyptus plantations and found that infiltration rates increased with stand age, which was attributed to enhanced preferential flow due to more abundant roots. Zhao et al. (2022), on the other hand, observed a decrease in infiltration rates with increasing forest age within the first 7 years of a Eucalyptus plantation. This decrease in infiltration rates was attributed to SWR induced by a higher SOM content, as well as to higher bulk density due to the compacting effect of the trees root system. Zema et al. (2021) found that $K(h)$ tended to increase with increasing pine forest age (within 4 categories: 1–19, 20–39, 60–79 and 80–100 years). This was attributed to higher SOM content and porosity, and also to lower SWR values. These authors found that only the younger forests (1–19 years) showed a slight SWR (measured in terms of water drop penetration time), while older plots were not water repellent. As regards soil water retention curve, although it has been stated that SWR may change water potential at a certain water content (Smettem et al., 2021) there are not many studies analyzing stand age on the soil water retention curve and thus on water retention properties.

As it has been shown, there is not a right away relation between stand age and SWR and hydraulic properties such as $K(h)$ and infiltration rates, as both increases and decreases in these properties have been reported. The effect of forest age on soil hydraulic properties may be complex, as higher SOM content and increased macro and micro-biota activity may lead to more structured soils, and thus higher infiltration rates, while SWR may counteract this effect (Zema et al., 2021). While positive correlations between SWR and SOM content have often been reported in literature (e.g., Martínez-Zavala & Jordán-López, 2009; Olorunfemi and Fasinmirin, 2017; Chenu et al., 2000; Kajiura et al., 2012), some studies show that a positive correlation may not be present (Chai et al., 2023; Hallett, 2007). This may be because SWR depends on many factors including texture, specific surface, and SOM composition (Zema et al., 2021). However, the relation between SOM composition and SWR has

rarely been studied and is still not fully understood (Majid et al., 2023). Chai et al. (2023) studied the relation between the content of SOM fractions extracted by different solvents and SWR, and found that while some hydrophobic fractions were positively correlated to SWR, more than half of the studied fractions were negatively correlated to SWR. Other studies report strong correlation between SWR and certain SOM fractions (e.g., extractable lipids) or specific compound families (e.g., alkanes, waxes, sterols, insoluble fatty acids) (Lozano et al., 2013a; Majid et al., 2023). Due to the complexity of SOM, finding the relation between SWR and SOM fractions that are relatively easy to obtain may give valuable information to understand and predict this phenomenon.

Different methods have been used to estimate SWR. Water drop penetration time is a commonly used method, though it may be time consuming and lead to experimental error due to water evaporation in soils with severe SWR (Santos et al., 2016). Tillman et al. (1989) proposed the use of the repellency index (RI) defined as the ratio between ethanol and water sorptivities (S_e and S_w) to estimate SWR. It has been proved that RI can successfully distinguish between repellent and non-repellent soils, as S_w approaches S_e in non-repellent soils (Tillman et al., 1989). Also, RI is a useful tool to quantify sub-critical SWR that has been used under different management and soil conditions (Hallett et al., 2004; Lipiec et al., 2009; Król et al., 2013; Ajayi & Horn, 2016). More recently, interest has been placed on estimates of SWR that are obtained from water infiltration curves alone (Alagna et al., 2017). This is the case of the Water repellency cessation time (WRCT) proposed by Lichner et al. (2013) and the Modified repellency index (Rim) (Sepehrnia et al., 2016). Using a different approach, Pachepsky & Karahan (2022) proposed the classification of cumulative infiltration curves according to their shape, which allows to qualitatively determine the effect of SWR. Comparing different methods to estimate SWR may help understand the nature of this phenomenon, as well as each method usefulness under particular field conditions. Furthermore, deepening knowledge of SWR temporal evolution and its relationship with SOM and hydraulic properties can help to develop more sustainable land management practices and to understand the effect of this phenomenon on hydrological processes. We hypothesized that: i- SWR increases with stand age, as SOM content increases and SOM composition changes; ii- hydraulic properties change with stand age, as a result of changes in pore system configuration; and iii- WRCT and Rim are able to evaluate SWR in Eucalyptus stands of different ages.

The aims of this study were: i- to determine the influence of Eucalyptus stands of different ages on soil physical and hydraulic properties and SWR phenomena; ii- to evaluate the effects of SOM content and composition on SWR; iii- to determine the relationship between SWR and soil physical properties; iv- to compare different methodologies for SWR evaluation.

2. Materials and methods

2.1. Site and treatments

The experimental plots are located in the “Ing. Agr. Julio Hirschhorn” experimental station belonging to the Faculty of Agro-nomical and Forestry Sciences (FCAYF), of the National University of La Plata (UNLP), La Plata, Argentina (34° 58' 56" S, 57° 59' 48" O). Climate in the region is warm temperate, fully humid (Cfa type according to the Köppen-Geiger climate classification) (Kottek et al., 2006) with a mean annual temperature of 16.4 °C and a mean annual rainfall of 1083 mm. The studied soil was classified as a Typic Argiudoll (UDSA classification) (Hurtado et al., 2006). The horizon sequences and their depths are: A (0–29 cm), AB (29–37 cm), Bt (37–115 cm), BC (115–160 cm) and Ck (160–200 cm) horizons (Soil Survey Staff, 2014). Soil properties in the A horizon are described in Table 1. Treatments consisted of four Eucalyptus stands of different age: i) 3 years old (2019–2022) *E. dunni* and *E. grandis* x *E. camaldulensis* plantation (3 yr) ii) 7 years old (2015–2022) *E. grandis* x *E. camaldulensis* plantation (7

Table 1
Main soil properties in the A horizon.

Texture	Depth (cm)	Silt (%)	Clay (%)	Sand (%)	Organic matter content (g kg ⁻¹)
silt loam	29–37	59.3	23.5	17.2	34.4

yr); iii) 11 years old (2011–2022) *E. dunnii* and *E. camaldulensis* plantation (11 yr); iv) 32 years old (1990–2022) *E. dunnii* plantation (32 yr).

2.2. Sampling

For each of the four treatments, ten undisturbed soil samples (5 cm height, 5 cm diameter) were taken from the top 5 cm of the soil (A horizon) (total number of samples = 40). Each undisturbed sample was used to determine bulk density and total porosity (TP), mini-infiltration, soil water retention curve, and sorptivity. In addition, two composite samples (disturbed soil) were taken from each treatment in order to determine specific surface (SS), total organic carbon (TOC), and different physical and chemical fractions of SOM (Total number of samples = 8). Samples were collected in the surface layer (0–5 cm) after manually removing organic litter, when present.

2.3. Soil properties determination

TOC content at the 0–5 cm layer was measured according to Walkley & Black (1934). Chemical fractionation was performed in order to obtain fulvic acids (FA), humic acids (HA), and humin (hum) fractions, following the protocol described by Benites et al. (2003) that is a modification of the method described by Schnitzer (1982). Physical fractionation according to Duval et al. (2018) allowed to obtain coarse particulate organic carbon (POC_c, >106 μm), fine particulate organic carbon (POC_f, 53–106 μm) and mineral associated organic carbon (MOC, <53 μm) fractions.

Bulk density was determined according to Hillel (1998), and TP values was calculated from bulk density. SS was determined by the etilenglicol monoetileter method described by Heilman et al. (1965) and Carter et al. (1986). A sandbox apparatus was used to determine water retention data at tension (h) = 0, −10, −30, −50, −70 and −100 cm and a pressure chamber to determine water retention data at h = −6000 cm. Bulk density was used to transform gravimetric to volumetric water contents. Water retention data was fitted with the van Genuchten (1980) model, employing the retention curve code (RETC) (van Genuchten et al., 1991) to obtain the water retention curve and calculate the following capacity soil physical quality indicators: macroporosity (θ_{ma}, equivalent radio > 30 μm), mesoporosity (θ_{me}, equivalent radio = 15–30 μm) and microporosity (θ_{mi}, equivalent radio < 15 μm), plant available water capacity (PAW), and field capacity (FC) according to Reynolds et al. (2009), as well as transmission porosity (P_{tr}, 0.005–0.5 μm) and storage porosity (P_{st}, 0.5–50 μm) (Bodner et al., 2023).

Mini-infiltration runs were performed at three values of h (−6 cm, −3 cm and 0 cm, applied in this order and in the same sample) as described by Soracco et al. (2019). The tension was fixed by rising the sample h cm above the water level, and every determination at each tension took approximately 3 min to reach the steady state. Cumulative infiltration was determined as the ratio between the infiltration volume and the disc area. Cumulative infiltration curves were used to derive hydraulic conductivity at 0, −3, and −6 cm h (K₀, K₃, K₆) (Ankeny et al., 1991). Total porosity connectivity (Cw_{TP}) was calculated from K₀ values and TP according to Lozano et al. (2013b).

After mini-infiltration, samples were air dried, and then Se and Sw were determined in the same samples. The methodology used in this work was described by Villarreal et al. (2017) as an improvement to the Leeds-Harrison et al. (1994) method. For Se and Sw determination, infiltration runs with ethanol and water were performed in each sample,

in this order and allowing the samples to dry for one week in a desiccator chamber (Villarreal et al., 2024). Water sorptivity (Sw) and ethanol sorptivity (Se) were calculated according to equation (1):

$$S_{w,e} = \sqrt{\frac{Q_{w,e}f}{4br}} \tag{1}$$

Where Qw and Qe are the infiltrated flow rates of water and ethanol, respectively, at steady state, r is the infiltration radius, f is the air filled porosity, and b is a shape parameter taken as 0.55 (White & Sully, 1987).

For each sample, the pair of estimated values Sw and Se were used to calculate the RI according to Tillman et al. (1998) (equation (2):

$$RI = 1.95 \frac{S_e}{S_w} \tag{2}$$

Cumulative infiltration was plotted against t^{0.5}, in order to calculate the WRCT as the intersection point between the two straight lines obtained after performing a linear regression to the initial and the later times of these curves (Lichner et al., 2013). The RIm was calculated as the ratio between initial time slope and later time slope of these linear regressions (Sepehrnia et al., 2016).

Also, cumulative water infiltration curves at 0 cm tension were classified into 12 categories according to Pachepsky & Karahan (2022) (Table 2). As curve types E, K and L are expected to be found in water-repellent soils (Pachepsky & Karahan, 2022), curves were separated into classical (A type), water-repellent (E, K and L types) and others.

2.4. Statistical analyses

One way ANOVAs were performed to determine the effect of treatment (stand age) on the studied soil properties. The variables with a non-normal distribution were transformed to their logarithmic values to achieve normality (K₀, K₃, K₆, and Cw_{TP}) prior to the statistical analysis. Fisher's (LSD) test was used to compare means (Sokal & Rohlf, 1995). Correlation analyses were performed in order to investigate the relationships between the hydraulic properties and SWR (Spearman's correlation). All the analyses were performed with INFOSTAT software (Di Rienzo et al., 2008), and the significance level was determined at p-value < 0.05. Results are expressed as mean ± their standard deviations.

3. Results

3.1. Soil organic matter and specific surface

Values of TOC and SOM fractions are shown in Table 3. TOC and most SOM fractions were not significantly affected by stand age

Table 2
Cumulative infiltration curve shapes according to the classification proposed by Pachepsky & Karahan (2022).

Shape type	Description
A	classic concave-to-linear shape.
B	concave-to-linear shape with a non-monotonous infiltration rate; the concave section is partially above the extrapolation of the linear section.
C	concave-to-linear shape with a non-monotonous infiltration rate.
D	similar to type B, but with no data in the early stage of infiltration.
E*	convex-to-linear shape with a monotonously increasing infiltration rate.
F	convex-to-linear shape with non-monotonous infiltration rate.
G	variable curvature-to-linear shape.
H	linear shape with constant infiltration rate.
I	linear shape with a break in infiltration rate.
J	concave shape with continuously decreasing infiltration rate.
K*	convex shape with continuously increasing infiltration rate.
L*	convex-to-concave shape with no linear sections.

* indicate expected curves in soil water repellent conditions.

Table 3

Soil organic matter content and composition in Eucalyptus stands of different ages.

	TOC (g.100 g ⁻¹)	POCc (g.100 g ⁻¹)	POCf (g.100 g ⁻¹)	MOC (g.100 g ⁻¹)	FA (g.100 g ⁻¹)	HA (g.100 g ⁻¹)	hum (g.100 g ⁻¹)
3 yr	1.7 ± 0.1	0.10 ± 0.06	0.17 ± 0.01	1.44 ± 0.07	0.26 ± 0.03	0.27 ± 0.06b	1.19 ± 0.03
7 yr	2.2 ± 0.3	0.20 ± 0.2	0.06 ± 0.02	1.9 ± 0.5	0.34 ± 0.06	0.27 ± 0.01b	1.6 ± 0.4
11 yr	2.36 ± 0.01	0.2 ± 0.1	0.3 ± 0.1	1.8 ± 0.2	0.29 ± 0.08	0.27 ± 0.09b	1.80 ± 0.02
32 yr	2.16 ± 0.09	0.12 ± 0.04	0.3 ± 0.1	1.7 ± 0.1	0.30 ± 0.03	0.51 ± 0.05 a	1.3 ± 0.1

3 yr = 3 years stand, 7 yr = 7 years stand, 11 yr = 11 years stand, 32 yr = 32 years stand. TOC = total organic carbon, POCc = coarse particulate organic carbon, POCf = fine particulate organic carbon, MOC = mineral associated organic carbon, FA = fulvic acids, HA = humic acids, hum = humins. Total number of samples = 8. Results are expressed as mean values followed by their standard deviations. Different letters indicate significant differences among treatments (LSD-Fisher, p-value < 0.05).

(ANOVA, p-value > 0.05). HA was the only SOM fraction affected by treatment, being significantly higher in 32 yr. SS was significantly higher in 11 yr and 32 yr (36 ± 2 and 43 ± 10 m²/g, respectively), than in 3 yr and 7 yr (18 ± 5 and 9 ± 7 m²/g, respectively).

3.2. Pore size distribution

From ANOVA results, most pore configuration indicators (TP, θ_{ma} , θ_{me} , θ_{mi} , PAW, FC and P_{st}) were significantly affected by stand age (p-value < 0.05). P_{tr} was the only indicator that was not significantly affected by stand age. Most changes in pore size distribution occurred between 3 and 7 years, as well as between 11 and 32 years (Table 4). In the first years, a reduction in TP was observed, mainly due to a reduction in θ_{me} , which also caused a significant reduction in PAW and P_{st} . At later stages (between 7 yr and 32 yr, but mainly between 11 yr and 32 yr) the opposite trend in TP was observed, with an increase in TP with stand age due to increases in θ_{mi} , and thus in FC. Also a reduction in θ_{ma} as observed between 7 yr and 11 yr. Mean water retention curves are shown in Fig. 1.

3.3. Hydraulic conductivity, porosity connectivity and water sorptivity

ANOVA results showed that K_0 , K_3 and K_6 were significantly affected by stand age. For each tension K decreased with stand age, following the order 3 yr > 7 yr > 11 yr > 32 yr (Fig. 2). The values of K_0 ranged between 0.005 and 12 cm h⁻¹, the values of K_3 between 0.003 and 1.2 cm h⁻¹ and K_6 between 0.002 and 0.4 cm h⁻¹. Cw_{TP} values followed the same order (3 yr > 7 yr > 11 yr > 32 yr), with the following values: 23 ± 9 cm h⁻¹ for 3 yr, 0.8 ± 0.8 cm h⁻¹ for 7 yr, 0.3 ± 0.6 cm h⁻¹ for 11 yr, and 0.008 ± 0.001 cm h⁻¹ for 32 yr. Sw was also significantly affected by stand age, and followed the order 3 yr > 7 yr > 11 yr = 32 yr (0.08 ± 0.02 m/s⁻⁰(-|-)⁻⁵ in 3 yr, 0.05 ± 0.01 m/s⁻⁰(-|-)⁻⁵ in the 7 yr, 0.010 ± 0.02 m/s⁻⁰(-|-)⁻⁵ in 11 yr, and 0.004 ± 0.001 m/s⁻⁰(-|-)⁻⁵ in 32 yr).

3.4. Soil water repellency

SWR indexes were significantly affected by stand age. RI was lower for 3 yr and 7 yr, with no significant differences among these two, and

then increased with stand age (Table 5). WRCT values showed the following order 7 yr = 32 yr > 3 yr = 11 yr, while RIm was higher for 11 yr, followed by the rest of the treatments with similar values among them. Fig. 3 illustrates the linear regression used to determine WRCT and RIm in a “hockey-stick-like” cumulative infiltration versus the squared root of time curve, which is characteristic for water-repellent soils.

With regards to the classification of the curves according to Pachepsky & Karahan (2022), the following curve types were observed: A (39 % of the curves), F (16 %), H (16 %), G (13 %), E (11 %), C (3 %) and D (3 %). Water repellent curves were only observed for 3 yr (Table 6), being E the only water repellent curve type present. In 7 yr and 11 yr, classical cumulative infiltration curves were predominant, while in 32 yr linear curves were the most frequent type.

3.5. Correlation analysis

Correlation matrix (Fig. 4) shows that most variables derived from water infiltration experiments were correlated. R values between 0 and 0.3 indicate weak correlation, values between 0.3 and 0.5 indicate moderate correlation, values between 0.5 and 0.7 indicate strong correlation, and values between 0.7 and 1 a very strong correlation. Hydraulic conductivity at different h were correlated among them, strongly correlated to storage porosity and very strongly correlated to Cw_{TP} . P_{tr} was not significantly correlated with the other indicators. Sw showed a very strong positive correlation with K(h) (correlation coefficient > 0.7) and to Cw_{TP} . Regarding SWR indicators, RI showed a very strong negative correlation with Sw, as the first one is proportional to the inverse of the second one ($RI = 1.95 Se/Sw$). On the other hand, WRCT and RIm showed mainly moderate negative correlation with Sw and K (h) and no significant correlation with RI.

4. Discussion

In this study, stand age had no effect on TOC and most fractions of SOM content. The only fraction affected by stand age was HA, which shows Eucalyptus plantations can increase relatively soluble SOM in the long term. This also implies that differences in SWR could not be

Table 4

Pore size distribution and capacity indicators in Eucalyptus stands of different ages.

	TP (cm ³ .cm ⁻³)	θ_{ma} (cm ³ .cm ⁻³)	θ_{me} (cm ³ .cm ⁻³)	θ_{mi} (cm ³ .cm ⁻³)	PAW (cm ³ .cm ⁻³)	FC (cm ³ .cm ⁻³)	P_{tr} (cm ³ .cm ⁻³)	P_{st} (cm ³ .cm ⁻³)
3 yr	0.503 ±0.004b	0.11 ±0.03 ab	0.03 ±0.01 a	0.36 ±0.02c	0.12 ±0.03 a	0.31 ±0.04c	0.04 ±0.03	0.25 ±0.07 a
7 yr	0.493 ±0.001 d	0.12 ±0.02 a	0.020 ±0.002b	0.35 ±0.01c	0.086 ± 0.009b	0.32 ±0.01c	0.06 ±0.02	0.18 ±0.02b
11 yr	0.500 ±0.004c	0.08 ±0.02c	0.017 ±0.002b	0.40 ±0.02b	0.08 ±0.01b	0.37 ±0.01b	0.03 ±0.02	0.16 ±0.02b
32 yr	0.632 ±0.009 a	0.09 ±0.02 bc	0.017 ±0.005b	0.52 ±0.02 a	0.08 ±0.02b	0.49 ±0.02 a	0.05 ±0.02	0.15 ±0.04b

3 yr = 3 years stand, 7 yr = 7 years stand, 11 yr = 11 years stand, 32 yr = 32 years stand. TP = total porosity, θ_{ma} = macroporosity, θ_{me} = mesoporosity, θ_{mi} = microporosity, PAW = plant available water, FC = field capacity, and P_{tr} = transmission porosity, and P_{st} = storage porosity. Total number of samples = 40. Results are expressed as mean values followed by their standard deviations. Different letters indicate significant differences among treatments (LSD-Fisher, p-value < 0.05).

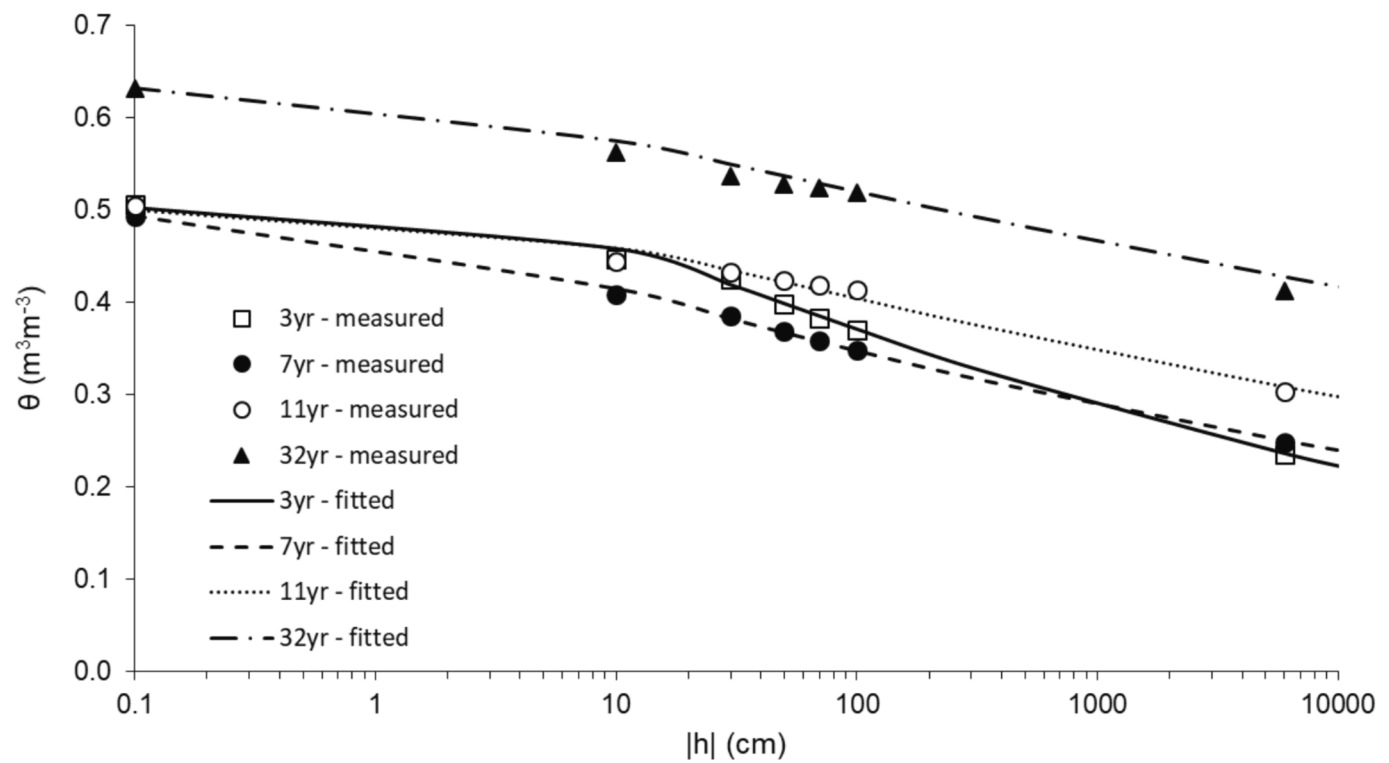


Fig. 1. Mean measured and fitted soil water retention curves in Eucalyptus stands of different ages. 3 yr = 3 years stand, 7 yr = 7 years stand, 11 yr = 11 years stand, 32 yr = 32 years stand. θ = water content; h = tension.

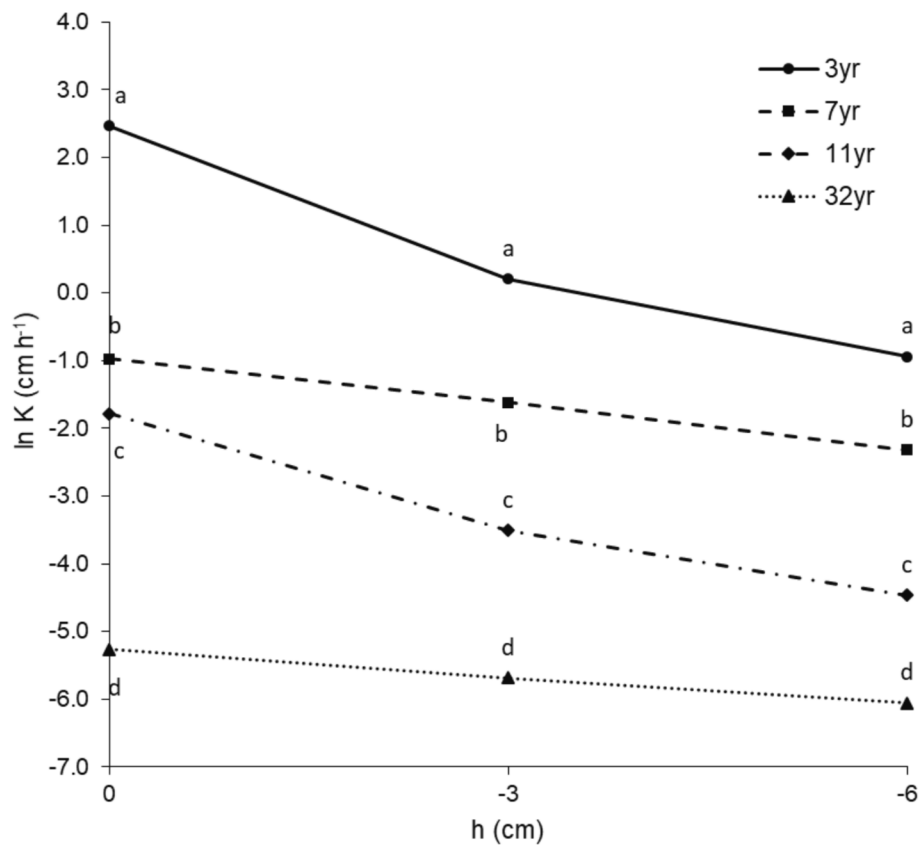


Fig. 2. Soil hydraulic conductivity (K) at 0, -3 and -6 cm tension (h) in Eucalyptus stands of different ages. 3 yr = 3 years stand, 7 yr = 7 years stand, 11 yr = 11 years stand, 32 yr = 32 years stand. For each tension, different letters indicate significant differences among treatments (LSD-Fisher, p -value < 0.05).

Table 5

Soil water repellency indicators in Eucalyptus stands of different ages.

	RI	WRCT (s)	RIm
3 yr	2.1 ± 0.4c	29 ± 15b	0.2 ± 0.2c
7 yr	3 ± 2c	87 ± 84 a	0.3 ± 0.3 bc
11 yr	20 ± 8b	43 ± 42b	2 ± 5 a
32 yr	52 ± 16 a	77 ± 27 a	0.4 ± 0.1b

3 yr = 3 years stand, 7 yr = 7 years stand, 11 yr = 11 years stand, 32 yr = 32 years stand. RI = repellency index, WRCT = water repellency cessation time, RIm = modified repellency index. Total number of samples = 40. Results are expressed as mean values followed by their standard deviations. For each parameter, different letters indicate significant differences among treatments (LSD-Fisher, p-value < 0.05).

explained by differences in SOM content and most SOM fractions. Similar results were reported by other authors (Kajiura et al., 2012).

Eucalyptus plantations caused a reduction in TP and θ_{me} within the initial 7 years, which may be the effect of soil compaction induced by more abundant roots in the first few years of the tree stand (Zhao et al., 2023). After this initial decrease, TP and θ_{mi} increased with stand age. This could not be attributed to higher SOM content, but can be explained by the structuring effect of roots and biota, and particularly of bacteria and fungi that stabilize soil structure at small scales (μm – cm) (Young et al., 1998). It is known that vegetation, and in this case replacing a grassland with a Eucalyptus plantation, affects soil microbial community as it causes changes in pH, moisture, nutrient balance, and SOM composition among other things (Lladó et al., 2018).

Hydraulic conductivity at different h decreased with stand age. The same happened with Sw, which describes the initial steps of the infiltration process. Zhao et al. (2022) observed a reduction in K(h) with stand age in a clay loam Oxisol, which was attributed to higher bulk density (due to the compacting effect of the roots) and to higher SWR.

Other authors observed the opposite effect in both eucalyptus and pine plantations, in a loam Oxisol and in a sandy clay loam Entisol (Zhao et al., 2023; Zema et al., 2021). In this study a reduction of K(h) could initially be related to a reduction in TP, but in the later stages an increase in TP occurred. A lack of relation between TP and K(h), which may be explained by changes on pore size distribution and porosity connectivity (Strudley et al., 2008; Villarreal et al., 2020; Polich et al., 2023). On the other hand, it has been stated that in forest soils the presence of canopy as well as hydrophobicity alter water processes, favoring processes such as preferential flow over tree stem and main roots, rather than matrix infiltration (Di Prima et al., 2023). This indicates that lower matrix hydraulic conductivity may not imply less water entering the soil in the water balance, and that other approaches may be necessary to study water balance in forests and tree plantations.

RI was the only SWR indicator that showed an increase with stand age as was expected. Values for 3 yr and 7 yr were similar to those found by other authors (Lipiec et al., 2009; Król et al., 2013; Ajayi & Horn, 2016) for agricultural soils, while values for 11 yr and 32 yr were one order of magnitude higher, and similar to those found by Alagna et al. (2017) in forest soils. WRCT and RIm showed a more erratic behavior. Also, these parameters had higher relative standard deviations than RI. Values of WRCT and RIm obtained were around one order of magnitude lower than those found by Alagna et al. (2017) for different forest soils. This fact rises the question of the strengths and limitations of the different methodologies employed to determine SWR. RIm and WRCT parameters have the advantage of being determined from a single infiltration run at $h = 0$ cm, that is a relatively easy and routine measurement. However, these values are estimated using both the initial steps of infiltration (when water movement is driven by capillarity and where SWR is expected to be expressed), and later times where steady state and soil saturation has been reached (and thus SWR cannot be expressed). In this study, both Sw and K_0 increased with stand age,

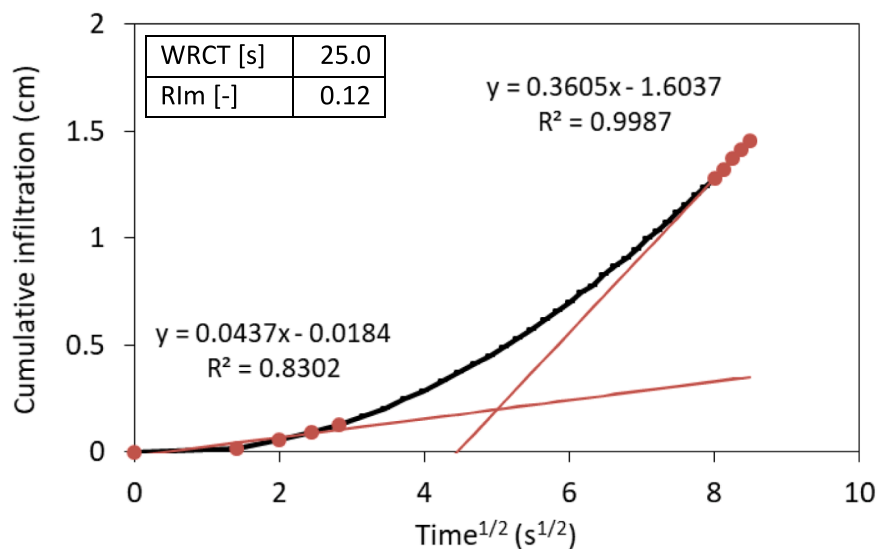


Fig. 3. “Hockey-stick-like” relationship of cumulative infiltration versus square root of time. The linear regressions for the determination of water repellency cessation time (WRCT) and the modified repellency index (RIm) are shown.

Table 6

Cumulative infiltration curves classes according to Pachepsky & Karahan (2022).

	Number of curves	Classical curves	Water repellent curves	Others	Predominant type
3 yr	10	1	4	5	E
7 yr	10	4	0	6	A
11 yr	9	7	0	2	A
32 yr	9	3	0	6	H

3 yr = 3 years stand, 7 yr = 7 years stand, 11 yr = 11 years stand, 32 yr = 32 years Eucalyptus stand.

disappeared or been minimized. Studying the relation between SWR and the difference between wetting and drying water retention curves may be an interesting topic for future researches.

5. Conclusions

In Eucalyptus stands of different age (3, 7, 11, and 32 years), Repellency index increases with stand age, independently of soil organic matter composition and content. Hydraulic conductivity decreases with stand age as a result of changes in pore system configuration (pore size distribution and porosity connectivity). Even though water repellency cessation time and the modified repellency index have the advantage of being determined from water infiltration curves alone, the shape of the curves should be taken into account to consider whether the use of these parameters is adequate. The repellency index is a more consistent soil water repellency parameter than water repellency cessation time and the modified repellency index. The kind of curves that may be found in water repellent soils should be reconsidered taking into account a larger number of samples with different soils and soil water repellency degrees, including for instance linear curves. Further research on the shapes of the cumulative infiltration curves in soils that exhibit water-repellency is encouraged, as well as on whether the determination of water repellency cessation time and the modified repellency index is possible in the different curves.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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